



STA 426 First Lecture

- survey
- course structure
- Some thoughts on Molecular Biology (Hubert)
- Exercise 1



Today's structure

9.00-9.45: Survey + Course Structure (Mark)

10.00-10.45: Introduction to Molecular Biology (Hubert's slides)

11.00-11.45: Quarto + Exercise 1

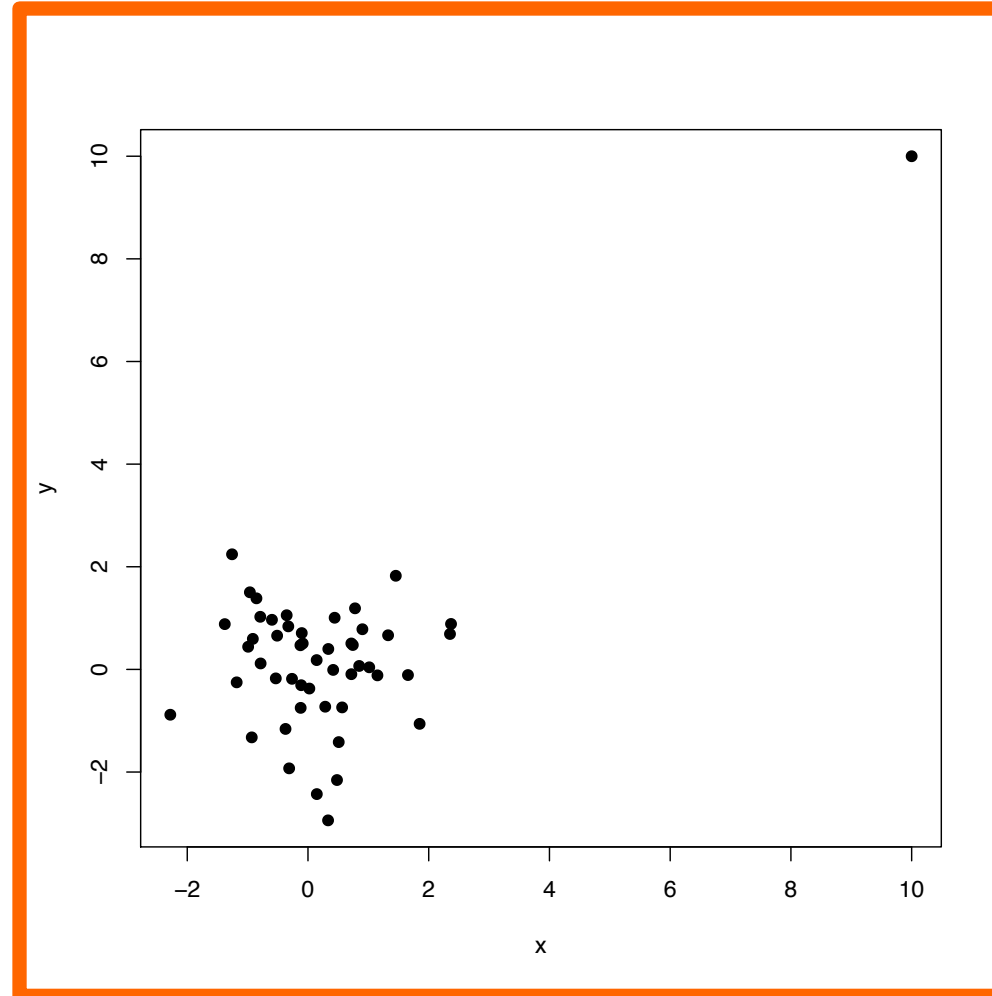


Survey: Statistical Insight

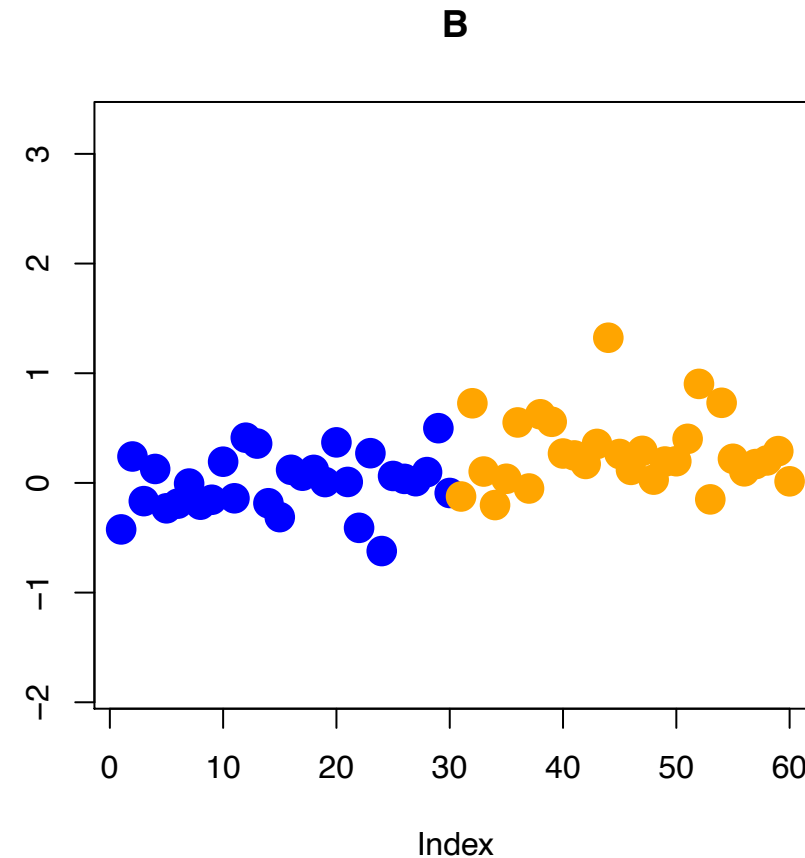
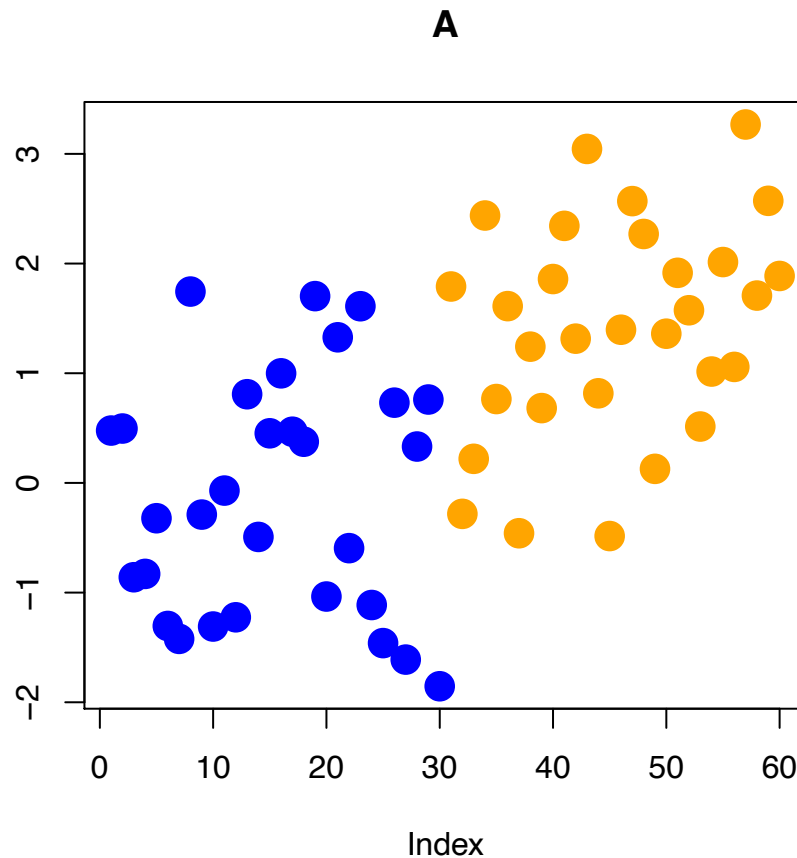




Question 1: In your view, what best describes the associations shown in the plot of 'x' and 'y' ?

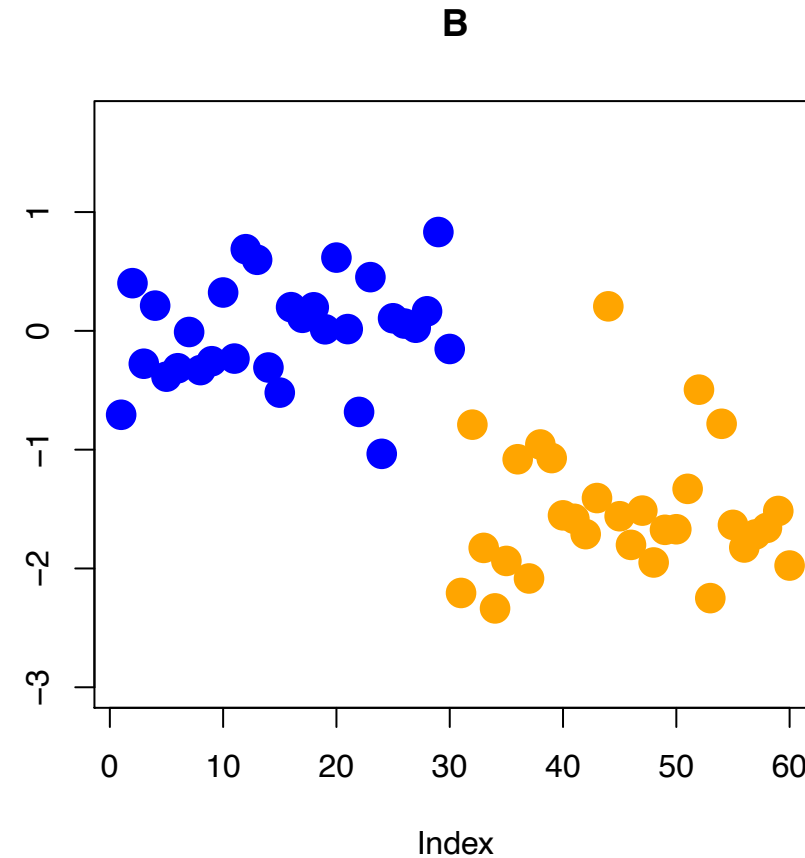
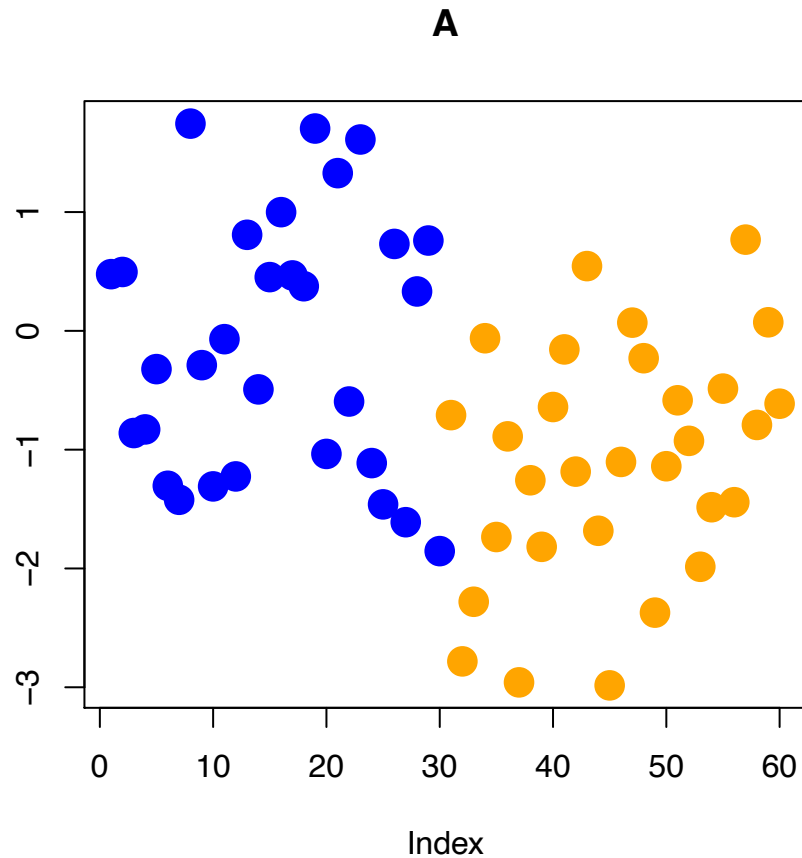


Question 2: Which plot highlights more (statistical) evidence for a change in the population means (between orange and blue)?





Question 4: Which plot highlights more (statistical) evidence for a change in the population means (between orange and blue)?

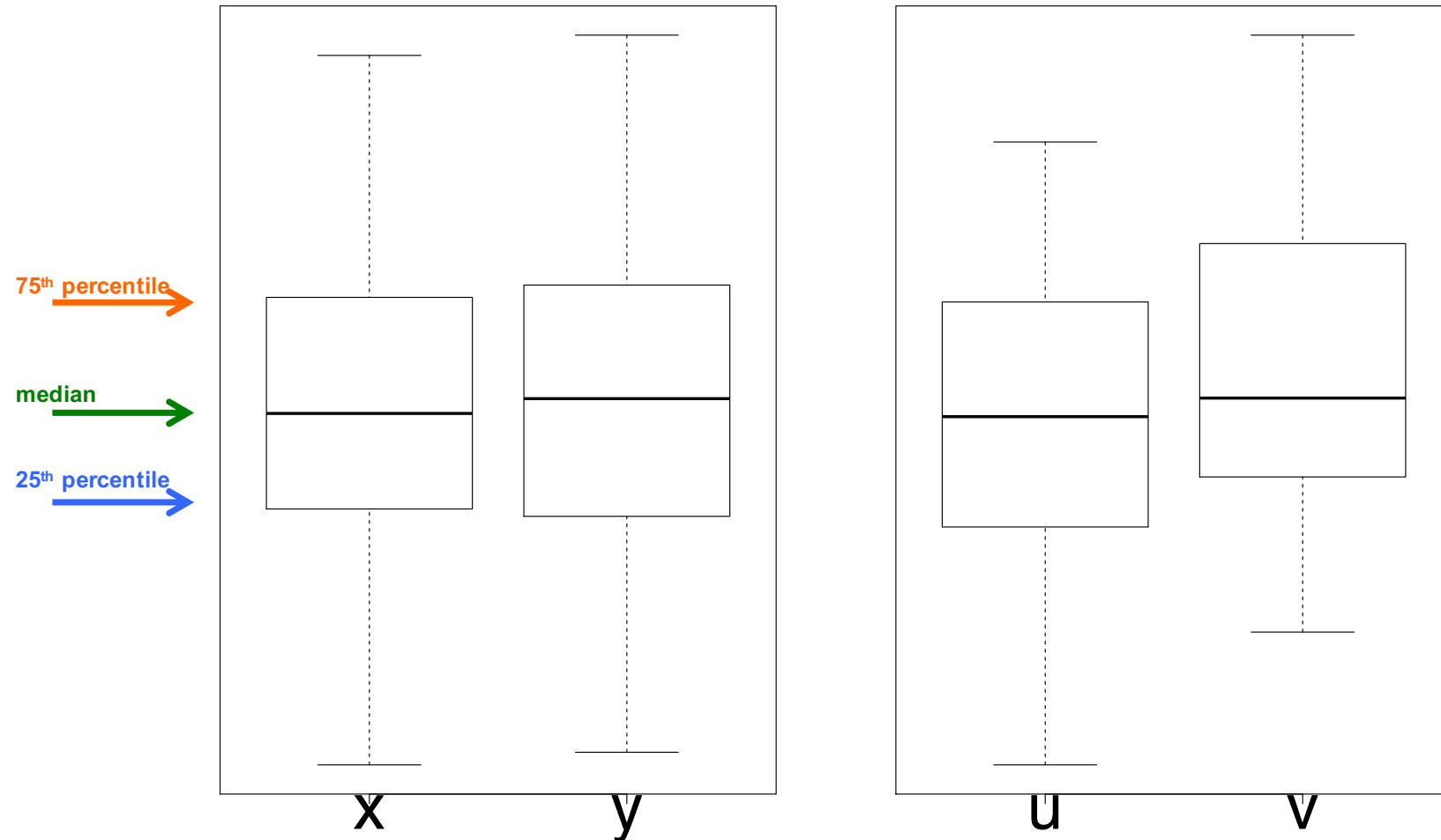




Question 5: Given this design matrix, describe the experimental design.

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Question 6: Given these boxplots, which of two underlying distributions are more similar?



Question 7: Of these equations, which one resembles the standard two sample t-test ?

1
$$\frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

2
$$\sum^k \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

3
$$\frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$



Course communication

- Video situation: we plan to have recordings, but no live stream
- Slack: all communication on Slack (note: all invitations were sent to UZH email addresses)
- Except for exceptional circumstances, **no emails please**; communicate on Slack
- Slack policy: unless really private, ask questions in a public channel (n.b: ***good questions get good answers***); use threads when relevant; good manners/behaviours are expected



Course evaluation

1. Journal club presentation	20%
2. Project	50%
3. Exercises	30%



Rough structure of lecture/exercise time

Monday mornings: we will run X.00-X.45; X in {9,10,11}

- Lectures and Exercises
- Lecture/journal club presentation (9.00-whenever)
- Remaining time: free (can be used to work on exercises; we are available for questions)

Critical skills needed by statisticians (Jeffrey Leek's words):

With all the excitement going on around statistics, there is also increasing diversity. It is increasingly hard to define “statistician” since the definition ranges from [very mathematical](#) to [very applied](#). An obvious question is: what are the most critical skills needed by statisticians?

So just for fun, I made up my list of the top 5 most critical skills for a statistician by my own definition. They are by necessity very general (I only gave myself 5).

1. **The ability to manipulate/organize/work with data on computers** - whether it is with excel, R, SAS, or Stata, to be a statistician you have to be able to work with data.
Python
2. **A knowledge of exploratory data analysis** - how to make plots, how to discover patterns with visualizations, how to explore assumptions
3. **Scientific/contextual knowledge** - at least enough to be able to abstract and formulate problems. This is what separates statisticians from mathematicians.
4. **Skills to distinguish true from false patterns** - whether with p-values, posterior probabilities, meaningful summary statistics, cross-validation or any other means.
5. **The ability to communicate results to people without math skills** - a key component of being a statistician is knowing how to explain math/plots/analyses.

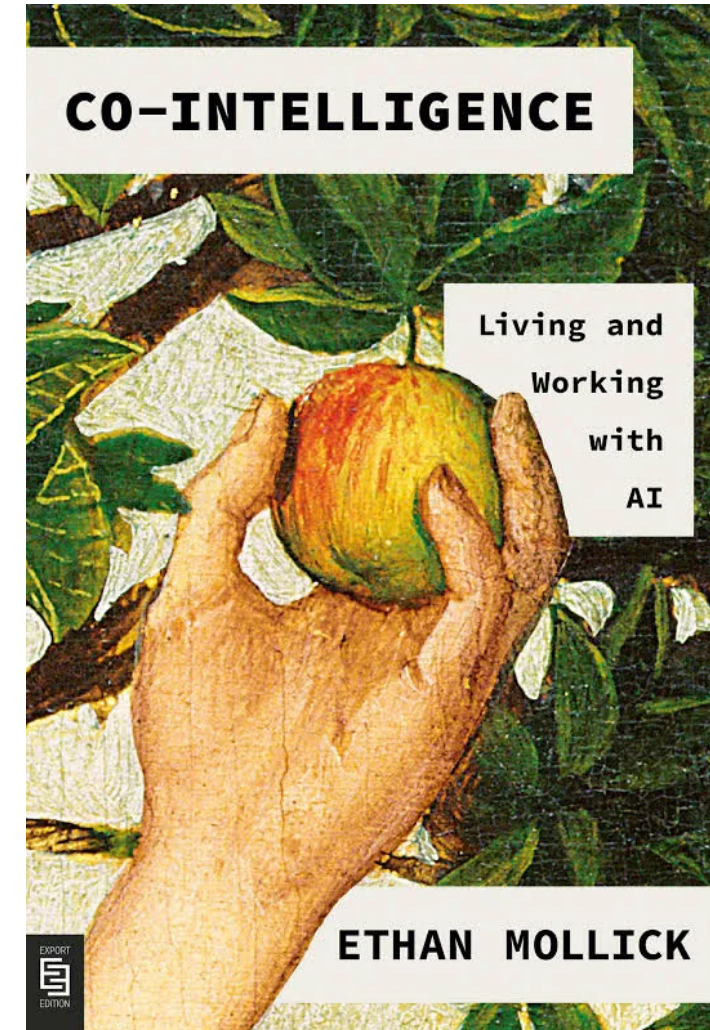


STA 426 Learning outcomes (in my words)

- Understand the fundamental “scientific process” in the field of Statistical Bioinformatics
- Be equipped with the skills / tools to preprocess genomic data (Unix, Bioconductor, mapping, etc.) and ensure reproducible research (R / markdown / quarto)
- Have a general knowledge of (some) **types** of data and **biological applications** encountered with high throughput genomic data
- Have the general knowledge of the range of statistical methods that get used with microarray and sequencing data
- Gain the ability to apply statistical methods / knowledge / software to a collaborative biological project
- Gain the ability to critical assess the statistical bioinformatics literature
- Write a coherent summary of a bioinformatics problem and it’s solution in statistical terms

The Co-Intelligence take on AI

- **Always invite AI to the table** --> Use AI frequently and experimentally. Understand its capabilities and limitations via hands-on use.
- **Be the human in the loop** -> You are responsible for judgment (ethics, verification, etc.). AI can generate convincing but incorrect and biased outputs. Think of AI as an assistant.
- **Treat AI like a person** --> AI responds better when given a role, context, and expectations. Remember it is not actually a person and lacks true understanding.
- **Assume this is the worst AI you will ever use** --> Capabilities are improving rapidly. Expect that future models will be much more capable. Collaborate with AI but remain adaptable.



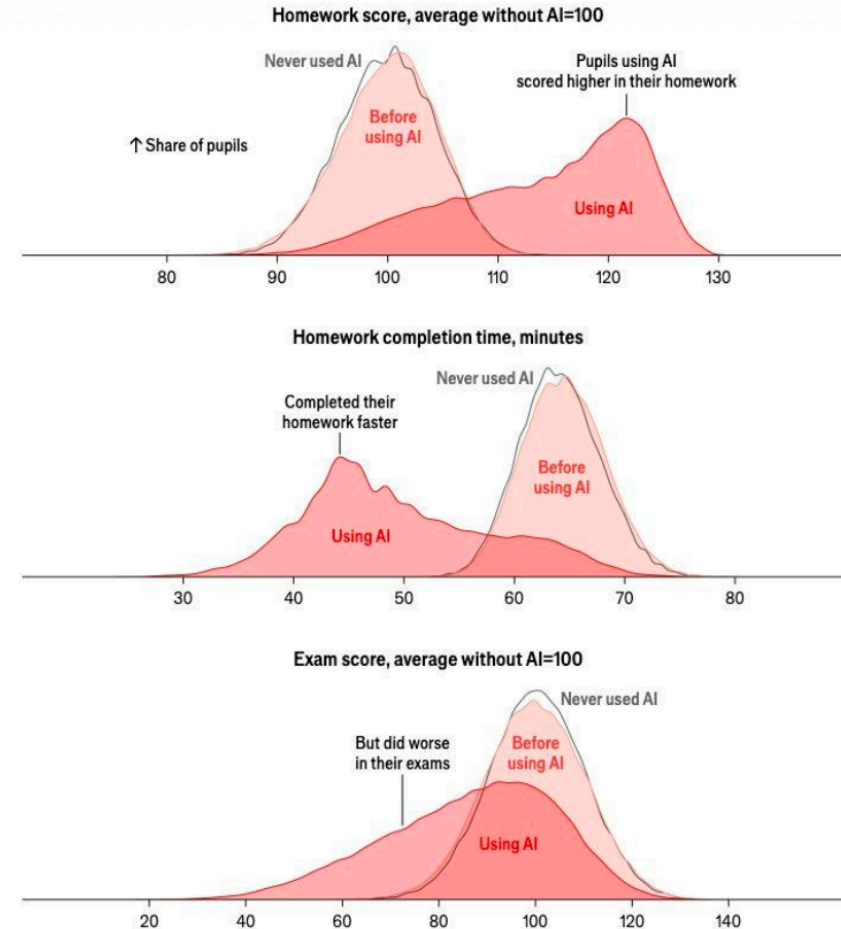
The Generative AI Learning Penalty

- AI adoption increases homework scores by **18%** and reduces completion time by **30%**.
- However, it lowers monthly exam scores by **20% within six months**.
- The full exam score penalty appears only after **about two years**.
- Learning losses occur primarily among students who **outsource homework to AI**.
- AI users who **spend about the same amount of time on homework** as non-users experience **only small learning losses**.

Stromberg, David and Lei, Victor and Wu, Yanhui, The Generative AI Learning Penalty: Evidence from Chinese Secondary Education (June 02, 2026). <http://dx.doi.org/10.2139/ssrn.6868618>

AI ate my homework

China*, effect of using AI, January 2023-June 2025



Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18



UZH has AI Guidelines / Policies

[https://www.informationsecurity.uzh.ch/dam/jcr:5c0a441a-369f-4090-bb29-3685d825e643/Guidelines%20for%20the%20Use%20of%20AI%20Systems%20\(AI%20Guidelines\).pdf](https://www.informationsecurity.uzh.ch/dam/jcr:5c0a441a-369f-4090-bb29-3685d825e643/Guidelines%20for%20the%20Use%20of%20AI%20Systems%20(AI%20Guidelines).pdf)

<https://www.uzh.ch/en/explore/basics/ai.html>

<https://www.uzh.ch/en/explore/basics/ai/recommendations.html>

<https://www.ai.uzh.ch/en/research/ai-policies.html>



AI draft policy (subject to change)

AI tools can and should be used for brainstorming and debugging assistance, but all submitted code must be written and understood by the student. You must explicitly cite all AI usage (e.g., in all exercises: section of template) in all presented / submitted work and explain how you verified the code (e.g., a README file or comments within the scripts/reports). *Copy-pasting from AI is considered academic dishonesty.* A 'human-in-the-loop' approach is required: you must be able to explain every line of code you submit.



The semester-long course structure (subject to change)

Date	Lecturer	Topic	Exercise	JC1	JC2
14.09.2026	Mark+Hubert	admin; mol. bio. basics	quarto; git(hub)		
21.09.2026	Mark	hands-on session #1: benchmarking	group exercise: Omnibenchmark		
28.09.2026	Hubert	NGS intro; exploratory data analysis	EDA in R		
05.10.2026	Hubert	mapping	Rsubread		
12.10.2026	Mark	limma + friends	linear model simulation + design matrices		
19.10.2026	Hubert	RNA-seq quantification	RSEM	X	X
26.10.2026	Mark	edgeR+friends 1	basic edgeR/voom	X	X
02.11.2026	Mark	edgeR+friends 2	advanced edgeR/voom	X	X
09.11.2026	Mark	hands-on session #2: RNA- seq	FASTQC/Salmon/etc.	X	X
16.11.2026	Hubert	single-cell 1: preprocessing, dim. red., clustering	clustering	X	X
23.11.2026	Mark	single-cell 2: clustering, marker gene DE	marker gene DE	X	X
30.11.2026	Mark	spatial omics 1	spatial statistics	X	X
07.12.2026	tba	hands-on session #3: single- cell RNA-seq	full scRNA-seq pipeline	X	X
14.12.2026	Mark	spatial omics 2	structures, DSP	X	X



Hands-on sessions

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02.11.2026	Mark	edgeR+friends 2	advanced edgeR/voom	X	X
09.11.2026	Mark	hands-on session #2: RNA- seq	FASTQC/Salmon/etc.	X	X
16.11.2026	Hubert	single-cell 1: preprocessing, dim. red., clustering	clustering	X	X
23.11.2026	Mark	single-cell 2: clustering, marker gene DE	marker gene DE	X	X
30.11.2026	Mark	spatial omics 1	spatial statistics	X	X
07.12.2026	tba	hands-on session #3: single- cell RNA-seq	full scRNA-seq pipeline	X	X
14.12.2026	Mark	spatial omics 2	structures, DSP	X	X



Expectations: **journal club** presentation

- 20-25 minutes (+5 minutes discussion)
- MUST:
 - ➔ be a paper about a **statistical** method in bioinformatics
 - ➔ be approved by Mark/Hubert
- Should:
 - ➔ describe the biological context and/or data collected
 - ➔ describe the (new) model used
 - ➔ describe comparisons to existing methods
- Should not:
 - ➔ be one of the papers discussed in detail in lectures: limma, edgeR, DEXSeq, etc.
- part of the JC grade: giving feedback to fellow students (online feedback forms)



Expectations: **project**

- ~10-15 page report, with R code in line (e.g. **quarto**)
- Describe the biological setting, statistical analysis, exploratory analysis with publication-quality graphics embedded
- 2026 theme:
 - ➡ Benchmarking / evaluation of computational tools
 - ➡ Omnibenchmark
 - ➡ Research-based learning
- Be strategic: work on something related to your interests!
- Typically due at end of first working week of January



Expectations: **exercises**

- There will be an exercise **every** week
- Across 14 weeks, there are 14 exercises
- The best 8 exercises are counted towards the 30% of the grade
- I recommend you also attempt the last exercises (spatial omics!)
- In general, exercises are due the following Tuesday (8 days after the lecture)



Soft technical skills needed (developed) in this course ...

- **Data Science!**

- Use unix-like operating system to run command-line programs
- if Windows, use cygwin / gitbash / WSL etc.)
- R: from the command line or RStudio / Posit (<https://posit.co/>); getting help; creating workflows; how to make publication-quality graphics (ggplot2); knitr / Rmarkdown / **quarto**
- Bioconductor – www.bioconductor.org
- Python + python environments
- git / GitHub / gitlab / CI-CD (GitHub Actions)
- bioconda / Docker / apptainer / podman (pick your poison)



Hubert lecture



Demos:

- Slack
- git / github
- quarto.org



Quick intro to Git/Github (version control)

```
git clone
git pull
git status
git branch
git checkout
git add
git commit
git push
```

<https://www.simplilearn.com/tutorials/git-tutorial/git-commands>



Exercise 1

Part A: GitHub setup. Accept the GitHub Classroom assignment. Clone the repository locally. In this cloned repo, create a `README.md` file containing the following: Name, GitHub username, Matriculation Number. Commit and push to your repo.

Part B: Quarto notebook. In R, use `data(penguins)` to retrieve the penguins dataset. Using the template (and naming it accordingly), create a quarto document that quickly explores the dataset, plots flipper length (`flipper_len`) versus body weight (`body_mass`), fits a linear regression, and gives a 1-2 sentence interpretation of the result. Check in both the QMD and HTML file to your repo.

Part C: GitHub Pull Request. A very typical mode of collaboration and code review via GitHub is branching and pull requests. Using your repository from Part A, create a branch called 'about-me', add a file called `firstname-surname.md` (use your actual name) with just STA426 name in the file. Open a pull request to the 'main' branch (leave it open; do not merge it)